

Erdős Problem #177

EP-177

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Find the smallest $h(d)$ such that the following holds. There exists a function $f : \mathbb{N} \rightarrow \{-1, 1\}$ such that, for every $d \geq 1$, $\max_{P_d} \left| \sum_{n \in P_d} f(n) \right| \leq h(d)$, where P_d ranges over all finite arithmetic progressions with common difference d .

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>